

Abstract

Arctic sea ice motion prediction is vital for maritime navigation as climate change impacts Arctic conditions. This work reproduces and extends the machine learning method from Hoffman et al. [2023] for daily sea ice motion forecasting. We implement four models: Persistence (PS), Linear Regression (LR), CNN, and a novel uncertainty-weighted CNN. Using wind velocity, sea ice concentration, and prior-day ice velocity, we predict current sea ice motion. Our main contribution is an uncertainty-aware CNN that reduces the influence of high-uncertainty observations—common in coastal areas—by weighting them less in training. This improves performance in hazardous coastal regions, despite modest overall metric gains (correlation 0.7955 vs. 0.8023 baseline). Limited overall improvement reflects the dominance of low-uncertainty areas, where coastal gains are a small part of the domain. This work highlights the value of integrating data quality into geophysical ML, especially for critical applications needing strong performance in challenging regions.

1 Introduction

The Arctic is a region known for its cold climate and large amounts of sea ice. Historically, this presence of thick sea ice has limited maritime activity in the Arctic. However, due to warmer temperature as a result of global warming, there has been a steep reduction in sea ice presence, which can create potential for new maritime activity in the region (Bennet and Smith [2020]).

However the change in the climate has also created a transition to thinner ice, which is much more fragile, and thus more prone to drifting. Drifting ice is hazardous to navigation, and creates uncertainty when it comes to ice flows in and out of specific micro regions.

In our project we aim to use Machine Learning to forecast daily sea ice movement, by reproducing the work presented in Hoffman et al. [2023].

Specifically we aim to use data driven machine learning using publicly available datasets of wind velocity and previous day sea ice concentration/velocity in order to make predictions of sea ice motion. We aim to recreate and evaluate the performance of 3 different modes:

- Persistence (PS)
- Linear Regression (LR)
- Convolutional Neural Network (CNN)

We also test our own novel CNN architecture that takes in parameter of uncertainty in order to try to improve upon already existing results. While both the LR and CNN models trained by Hoffman et al. [2023] performed well, overall, as novel sea ice forecasters, several limitations remain. The models do not explicitly account for the inherent uncertainty in the satellite-derived sea ice observations, which can vary significantly across different regions and seasons. As climate change continues to alter

Arctic conditions, sea ice is becoming increasingly dynamic and unpredictable, with rapid changes in ice extent, thickness, and motion patterns.

To address this gap, we introduce an uncertainty-weighted loss function that leverages the error estimates provided with the sea ice motion dataset. Our approach implements inverse uncertainty weighting, where observations with higher uncertainty contribute less to the training loss, while high-confidence observations receive greater emphasis. This allows the model to focus learning on the most reliable data points while appropriately down-weighting unreliable observations in problematic regions. Through this uncertainty-aware training framework, we hypothesize that the model will achieve improved generalization performance, particularly in challenging coastal regions where observational uncertainty is typically highest, ultimately leading to more robust predictions as Arctic sea ice continues to evolve under changing climate conditions.

2 Related work

Sea ice motion follows a momentum balance that includes atmospheric drag (τ_a), ocean drag (τ_w), Coriolis force ($m f \hat{k} \times \mathbf{u}$), ocean surface tilt ($mg \nabla H$), and internal ice stress ($\nabla \cdot \sigma$). Traditional models like CICE numerically solve this equation, but are computationally costly and sensitive to uncertain parameters, limiting operational reliability. Data-driven approaches directly map atmospheric/oceanic inputs to ice motion. Zhai and Bitz [2021] used a CNN with prior-day ice concentration, motion, and wind data, outperforming traditional models for daily forecasts. However, their model lacks explicit uncertainty handling and long-term dynamics. Park et al. [2024] proposed "Unicorn," combining U-Net and neural ODEs for weekly concentration forecasting, but it doesn't target motion vectors or temporal robustness. We reproduce Hoffman et al.'s CNN framework and extend it with a custom CNN that stacks temporal wind and ice features and predicts uncertainty, aiming for more robust Arctic sea ice forecasts.

3 Methodology

In our project, we use 4 ML and statistical based models to predict sea ice velocity components in the present day. These quantities are denoted

$$\begin{aligned} u_{i,t} \\ v_{i,t} \end{aligned}$$

. We specifically use a persistence based model (PS), a Linear Regression based model (LR), a CNN based model, and a custom CNN architecture variation, and compare their predictive capabilities. The first 3 architectures are reproduced from Hoffman et al. [2023], while the final architecture is our own custom variation that is intended to improve predictive capability by taking in uncertainty based parameters as input.

Input Data

The input to our models consists of the following parameters:

- Present day Zonal Wind Velocity ($u_{a,t}$) and Meridional Wind Velocity ($v_{a,t}$)
- Previous day Zonal Wind Velocity & Meridional Wind Velocity ($u_{i,t-1}, v_{i,t-1}$)
- previous day sea ice concentration (c_{t-1})

We choose not to include several parameters from the sea ice equation from Section 2; these are due to prior works stating that these parameters do not greatly impact the skill and correlation of predictions. Specifically, we see that sea ice thickness is not used as an input to the model, despite its physical relevance due to high uncertainty, and unavailability of data during the summer.

3.1 Baseline Reproduced Models

3.1.1 Persistence (PS)

The persistence model which we denote PS, simply predicts the present day velocity of sea ice $u_{i,t}^*$ to be equal to that of the prior day, $u_{i,t-1}^*$. That is:

$$u_{i,t}^* = u_{i,t-1}^*$$

This assumption offers us a baseline measure of the system variance and provides a lower bound on the skill that other models should reach.

3.1.2 Linear Regression

In the Linear Regression model, we want to regress each of our 5 input parameters onto sea ice velocity components.

To train the model, we simply solve the following regression for parameters, A , B , C , and D .

$$u_{i,t}^* = Au_{a,t}^* + Bu_{i,t-1}^* + Cc_{i,t-1}^* + D$$

u_i^* , u_a^* , c_i^* are all complex valued variables, with c_i having imaginary part set to 0.

Our goal is to solve the optimization problem, which has a loss function defined by:

$$\min_d \|Gd - u_i\|_2^2$$

where $d = [D \ A \ B \ C]^T$ and $G = \begin{bmatrix} 1 & u_{a,t}^* & u_{i,t-1}^* & c_{i,t-1}^* \end{bmatrix}$ u_i is the true sea ice velocity.

We perform our linear regression **gridwise**, which means we regress over the input parameters for each separate point on the grid of the region. This means that we compute a separate A to D for each specific grid point, creating hyper localized regression coefficients. Since regression of this format can easily blow up, we introduce a parameter λ that is used to limit the magnitude of the regression coefficients. Thus, our optimization problem becomes

$$\min_d \|Gd - u_i\|_2^2 + \lambda I$$

We pick the value of λ by using an iterative approach presented in Marquardt and Snee [1975], which gives us $\lambda = 10^{-2}$. We use the learned values of d to be our parameters to use on testing data, such that we simply set the sea ice velocity for that day

$$u_{i,t}^* = Gd$$

We evaluate the model using simple metrics that are further specified in the **Evaluation Metrics** section of this paper.

3.1.3 Convolutional Neural Network (CNN)

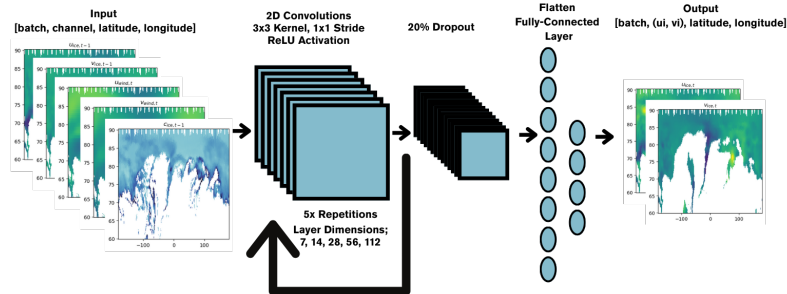


Figure 1: Baseline CNN architecture

The CNN model is designed to learn spatial features from gridded geophysical inputs in order to predict sea ice motion. Specifically, the network ingests gridded inputs of zonal wind velocity u_w , meridional wind velocity v_w at the present time t , and, sea ice zonal velocity u_i , meridional ice

velocity v_i , and sea ice concentration c_i , all at time $t - 1$, over a spatial domain $\Omega \subset \mathbb{R}^2$ discretized into a grid of size $H \times W$.

Let $\mathbf{X} \in \mathbb{R}^{5 \times H \times W}$ denote the input tensor at each timestep, where the three channels correspond to u_a , u_i , and c_i . The CNN learns a nonlinear mapping:

$$f_\theta : \mathbb{R}^{5 \times H \times W} \rightarrow \mathbb{R}^{2 \times d}$$

where d is the dimension of the flattened latent feature vector, and θ denotes all trainable parameters of the CNN. This mapping gives us output values of the meridional and zonal sea ice velocity forecast.

The CNN for this model is set up with 5 repetitions of a block unit. This block unit consists of the following structure, a 2D Convolution, a ReLU, and max pooling. This is followed by a 20% dropout layer, a flattening to 1D layer, and finally a regression onto a 1D vector of the output prediction. We then concatenate this output onto 2 maps of meridional and zonal sea ice velocity. To train the model, we use the Normalized Root Mean Squared Error (NRMSE) as the loss function, which penalizes large prediction errors while accounting for the scale of the target values.

$$\text{NRMSE} = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{y}_i - y_i)^2}}{y_{\max} - y_{\min}} \quad (1)$$

where $\hat{y}_i$ is the model prediction for the i -th sample, y_i is the ground truth value for the i -th sample, n is the number of samples, $y_{\max}$ and $y_{\min}$ are the maximum and minimum values of the ground truth vector $\{y_i\}_{i=1}^n$.

3.2 Proposed Improvement

3.2.1 Weighted CNN NRMSE Loss

One major issue with the baseline CNN comes from the highly variable uncertainty in the ground truth ice velocity vector data used in training. Ice velocity vectors are difficult to record directly, and so the Polar Pathfinder product gives estimates based on multiple input sources and interpolation. We cannot entirely trust that the trained model is providing reliable predictions uniformly throughout the region across all seasons.

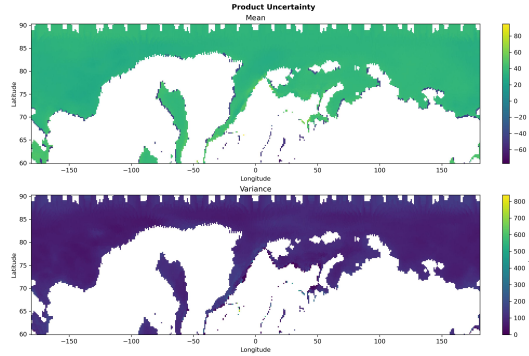


Figure 2: NSIDC Polar Pathfinder ice velocity uncertainty estimates.

Therefore, we propose a penalized loss function, where the baseline CNN NRMSE is weighted by the uncertainty in the ground-truth data (with a unique uncertainty value for each spatial grid point across time), given by;

$$\text{Weighted NRMSE} = \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n [(\hat{y}_i - y_i)^2 / r]}}{y_{\max} - y_{\min}} \quad (2)$$

Our goal is to penalize the training points with high uncertainty, so that we might produce a model that better represents the 'true' sea ice velocity. In comparing the metrics to those of the baseline CNN, and studying differences in those metrics across space and time, we hope to gain a better understanding of how model skill is truly affected by environmental and seasonal variations.

3.3 Evaluation Metrics and Baseline

In evaluating our ice motion prediction model, we use four key metrics: **RMSE**, **NSE**, **Correlation**, and **Skill**, each highlighting different aspects of performance. The model was trained to minimize **Root Mean Square Error (RMSE)**, which measures the average size of prediction errors and penalizes large errors, ensuring predicted ice velocities are close to the true values. Since RMSE alone does not reflect how well the model captures spatial or temporal patterns, we include other metrics. The **Nash–Sutcliffe Efficiency (NSE)** compares model performance to a simple mean predictor, with values near 1 indicating better skill. The **Correlation** coefficient measures how well predicted and observed ice motions vary together over space or time. Finally, the **Skill** score normalizes RMSE by the variability in observations, showing how much of the observed variation the model explains. Together, these metrics provide a balanced view: RMSE and Skill focus on error magnitude, while Correlation and NSE assess the model’s ability to capture the underlying structure and variability of ice motion.

Root Mean Squared Error (RMSE)

We define RMSE as a measure of the average magnitude of the prediction error:

$$\text{RMSE}_{x,y} = \sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2}$$

where, x_i is the ground truth (true) value at index i , y_i is the predicted value at index i , and n is the total number of samples.

Skill Score

We define:

$$\text{Skill}_{x,y} = 1 - \frac{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - y_i)^2}}{\sqrt{\frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})^2}}$$

where, $\bar{x}$ is the mean of the ground truth values x_i , The numerator is the RMSE of the prediction, The denominator is the standard deviation of the true data.

Nash–Sutcliffe Efficiency (NSE)

We define:

$$\text{NSE}_{x,y} = 1 - \frac{\sum_{i=1}^n (x_i - y_i)^2}{\sum_{i=1}^n (x_i - \bar{x})^2}$$

Correlation Coefficient

We define correlation as:

$$\text{corr}_{x,y} = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}}$$

where: $\bar{x}$ and $\bar{y}$ are the means of the true and predicted values respectively.

4 Experiments

4.1 Dataset and Setup

4.1.1 Setup Parameters

- **Train/Val/Test Split:** We **train** on input years 1992-2016, **validate** on 2017-2018, and **test** on 2019-2020. This chronological split is essential for time series based forecasting problems, as it prevents leakage from future to past.
- **Input:** Present-day wind vector components ($u_{a,t}$, $v_{a,t}$), previous-day ice velocity vector components ($u_{i,t-1}$, $v_{i,t-1}$) and previous day ice concentration ($c_{i,t-1}$). All input parameters are re-grid from their respective spatial grids to the same regular latitude and longitude grid with approximately 0.25 degree resolution using nearest neighbor interpolation. The grid-wise temporal means are then removed from each respective input variable. Finally, each input variable is normalized by the respective global standard deviation.

- Output: Normalized Zonal and Meridional components of present-day sea ice velocity vectors. The outputs are multiplied by the standard deviations removed earlier, and added by the means - giving final predictions in centimeters per second.
- Metrics: We evaluate using NSE, Correlation, Skill, and RMSE evaluation metrics as defined in the **Evaluation Metrics and Baseline** section.

4.2 Results

Model	RMSE	Skill	NSE	Correlation
Persistence	0.5038	0.1724	0.3151	0.6576
Linear Regression	0.9832	-1.7342	-6.476	0.3703
CNN Baseline	0.5603	0.4013	0.6146	0.8023
Ours (Uncertainty-CNN)	0.5676	0.3935	0.6322	0.7955

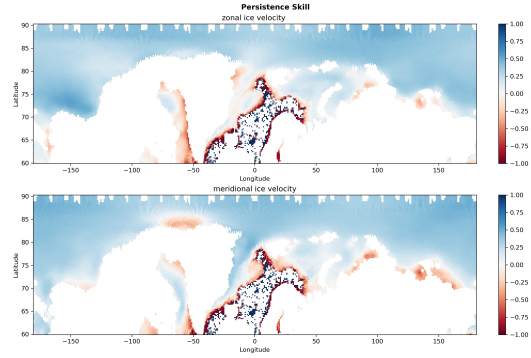


Figure 3: Baseline model skill.

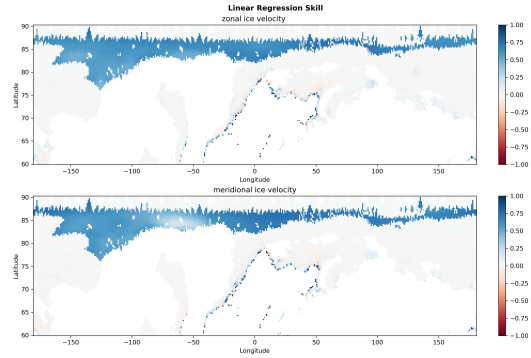


Figure 4: Linear regression skill.

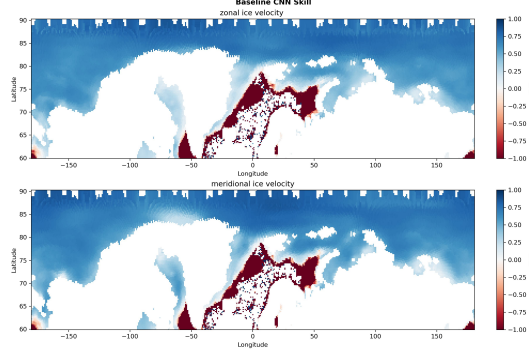


Figure 5: Baseline CNN skill.

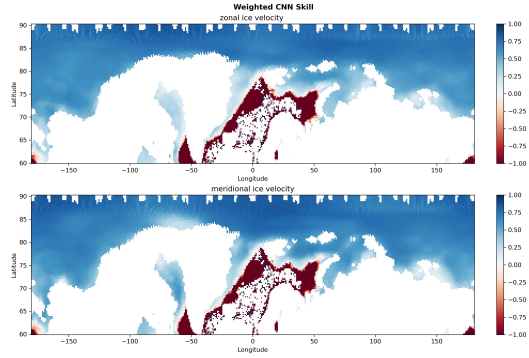


Figure 6: Weighted CNN skill.

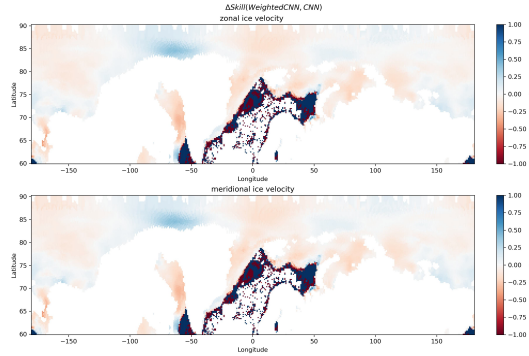


Figure 7: Difference in skill between weighted and unweighted baseline CNN.

4.3 Discussion

When trained on the entire training split, we see that the linear regression does not perform across the entire spatial domain. On this large data set, we see that input areas below 75 degrees latitude, where ice coverage might be sparse - we have a singular feature matrix due to similar inputs of minimal ice movement across time. Further work should investigate the de-correlation scales of the input data in order to restrict the training domain to the linear regression.

We see that both CNNs, however, perform across the entire spatial domain - as the data are input as images for each channel and time step. Globally, we see that the CNNs have greater error than the persistence baseline. This could be due to the fact that the CNN is underfitting and producing lower-magnitude predictions than the ground truth, overall (see time series in Additional Figures).

Spatially, we see that the CNN’s perform as expected with moderately uniform skill throughout the open-ocean regions in higher latitudes - where ice is free to move by the wind and less likely to be impeded by additional features. We see lower performance along coastlines, where ice tends to experience more variable periods of convergence and divergence - which could impede or promote ice motion. Ice movement here could also be influenced by unpredictable eddy currents off the coast.

The area of extremely poor skill, between Greenland and Scandinavia, is resultant of very sparse and seasonal sea ice coverage in that region. It’s likely that the ground-truth data in that region is not representative of actual sea ice velocity, and so the model is making predictions based on input parameters that don’t line up well with the targets.

When we observe the differences in skill between the weighted and unweighted CNN models, we see that there are distinct spatial patterns of performance differences that might suggest areas that are problematic in the CNN training, when uncertainties are not accounted for. We see significant improvements off the North Coast of Greenland for both the zonal and meridional components - suggesting that previous predictions were made worse by poor ground-truth data. We see subtle model improvements along some coastlines in the region, where ice velocity ground truth data is difficult to estimate. We see that model performance is worse in some open ocean areas which could suggest potential downfalls in the original training. Poor performance in the Russian Far North could suggest further influences of circulation and coastal ice convergence on model performance.

5 Conclusion

This project reproduced the machine learning method from Hoffman et al. (2023) for Arctic sea ice motion forecasting and introduced a novel uncertainty-aware CNN. We implemented four models—Persistence, Linear Regression, CNN, and uncertainty-weighted CNN—achieving strong baseline correlations up to 0.8023.

Our uncertainty-weighted CNN showed the value of including data quality metrics. While overall improvements were modest (correlation 0.7955 vs 0.8023), gains were notable in high-uncertainty coastal regions where traditional models struggle. The limited overall change reflects the dominance of low-uncertainty areas in the dataset. **Key contributions** include reproducing sea ice forecasting models and developing an uncertainty-aware training framework that better captures true sea ice velocity. **Future work** should explore advanced uncertainty methods, physics-informed losses, and validation on independent data. With increasing Arctic maritime activity, robust models focused on challenging coastal regions are critical for safe navigation and climate monitoring. This work lays a foundation for adaptable machine learning in the evolving Arctic environment.

Appendix

Github Link

<https://github.com/jbassham24/SeaIceECE228>

Team Member Contribution

Ariel: Ariel took up the task of doing the LR model, writing most of the report, and preparing poster.

Completed Evaluations

PLEASE NOTE
You have completed the following evaluations. You may edit and re-submit them before the deadline listed below.

Evaluation Type	Course	Deadline	
Instructional Assistant Student Evaluation of Teaching	SP25 ECE 228 - ML for Physical Applications (Shi, Yuanyuan)	6/7/2025 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	SP25 ECE 228 - ML for Physical Applications (Shi, Yuanyuan)	6/7/2025 8:00 AM	EVALUATE
Instructional Assistant Student Evaluation of Teaching	SP25 ECE 285 - Spec Topic/Signal&Image/Robotic (Antipa, Nicholas A)	6/7/2025 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	SP25 ECE 285 - Spec Topic/Signal&Image/Robotic (Antipa, Nicholas A)	6/7/2025 8:00 AM	EVALUATE
Instructional Assistant Student Evaluation of Teaching	SP25 ECE 287 - Spec Topic/Signal&Image/Robotic (Antipa, Nicholas A)	6/7/2025 8:00 AM	EVALUATE

(Ariel's eval)

Jack: Jack took up writing the CNN model, as well as processing the input data and running all our models on the full 27GB dataset of all years of data on his lab's servers and generated plots. He also contributed to poster/paper.

Completed Evaluations

PLEASE NOTE
You have completed the following evaluations. You may edit and re-submit them before the deadline listed below.

Evaluation Type	Course	Deadline	
Instructional Assistant Student Evaluation of Teaching	SP25 SHOC 228 - ML for Physical Applications (Shi, Yuanyuan)	6/7/2025 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	SP25 SHOC 228 - ML for Physical Applications (Shi, Yuanyuan)	6/7/2025 8:00 AM	EVALUATE

Figure 8: Jack Eval Confirmation

Alex: Alex took up doing the evaluation metrics for all our models. He also contributed to the poster and paper.

PLEASE NOTE
You have completed the following evaluations. You may edit and re-submit them before the deadline listed below.

Evaluation Type	Course	Deadline	
Graduate Course Student Evaluation of Teaching	SP25 ECE 228 - ML for Physical Applications (Shi, Yuanyuan)	6/7/2025 8:00 AM	EVALUATE
Graduate Course Student Evaluation of Teaching	SP25 ECE 285 - Spec Topic/Signal&Image/Robotic (Xie, Pengtao)	6/7/2025 8:00 AM	EVALUATE

Alex's Eval

Kaylan: Kaylan took up writing the custom CNN architecture. He also contributed to the paper.

✓ PLEASE NOTE

You have completed the following evaluations. You may edit and re-submit them before the deadline listed below.

Evaluation Type	Course	Deadline
Instructional Assistant Student Evaluation of Teaching	SP25 ECE 228 - ML for Physical Applications (Shi, Yuanyuan)	⌚ 6/7/2025 8:00 AM
Graduate Course Student Evaluation of Teaching	SP25 ECE 228 - ML for Physical Applications (Shi, Yuanyuan)	⌚ 6/7/2025 8:00 AM

(Kaylan Eval)

References

- A. Bennet and J. Smith. Arctic ice trends and impacts. *Journal of Arctic Research*, 2020.
- L. Hoffman, M. Mazloff, S. Gille, D. Giglio, C. Bitz, P. Heimbach, and K. Matsuyoshi. Machine learning for daily forecasts of arctic sea ice motion: An attribution assessment of model predictive skill. *American Meteorological Society*, 2023.
- D. W. Marquardt and R. D. Snee. Ridge regression in practice. *The American Statistician*, 29(1): 3–20, 1975. doi: 10.2307/2683673.
- J. Park, S. Hong, Y. Cho, and J. Jeon. Unicorn: U-net for sea ice forecasting with convolutional neural ordinary differential equations. *arXiv preprint arXiv:2405.03999*, 2024.
- J. Zhai and C. M. Bitz. A machine learning model of arctic sea ice motions. *arXiv preprint arXiv:2108.10925*, 2021.

Additional Figures

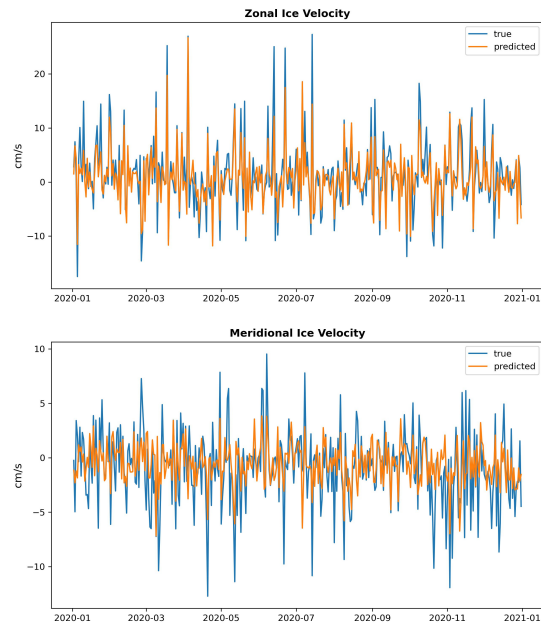


Figure 9: One year of CNN predictions at a specific location.

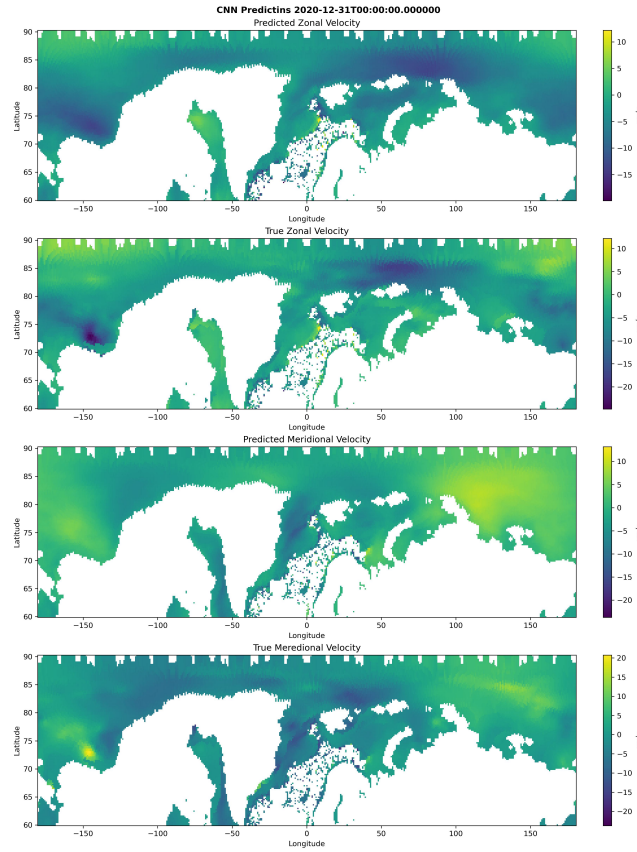


Figure 10: One day of True and CNN Predicted Velocities.

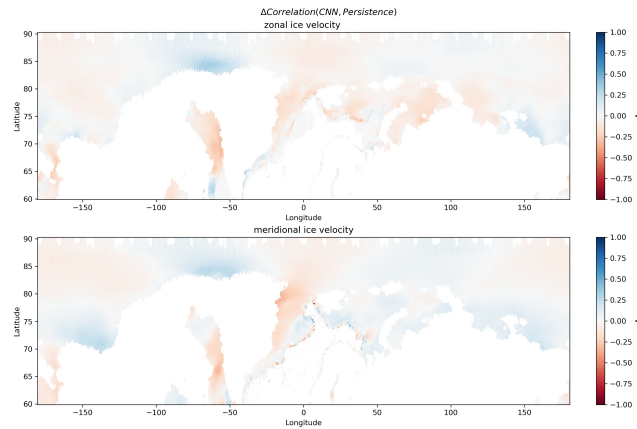


Figure 11: Difference in correlation between CNN and Persistence.

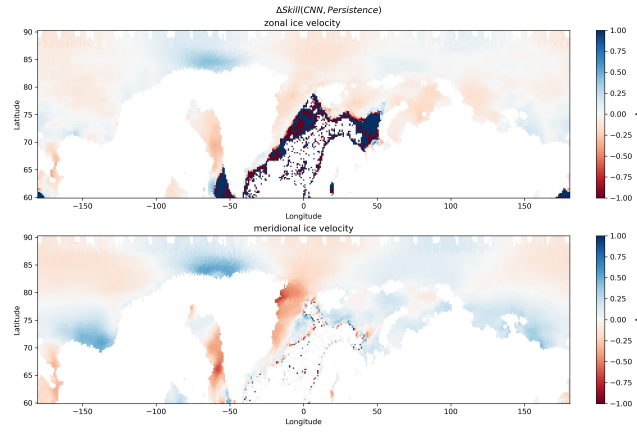


Figure 12: Difference in skill between CNN and and paper. Alex's Eval Kaylan: Kaylan took up writing the custom CNN architecture. He also contributed to the paper. 10 Persistence.